Return, Risk, and the Security Market Line

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Session Outline

13.1 Expected Returns and Variances

Expected Return ($E[R]$)

The weighted average of possible returns, where weights are the probabilities.

$$E[R] = \sum_{i=1}^{n} P_i \times R_i$$

Variance (σ^2) and Standard Deviation (σ)

Measures the volatility/spread of returns around the mean.

$$\sigma^2 = \sum_{i=1}^{n} P_i \times (R_i - E[R])^2 \quad ; \quad \sigma = \sqrt{\sigma^2}$$

Example: Calculating Expected Return

Scenario: Investment of **Rp100.000.000** in an Indonesian infrastructure stock.

State of Economy	**Probability (P_i)**	**Return (R_i)**
Recession	0,25	-10%
Normal	0,50	15%
Boom	0,25	35%

Calculation:

- $E[R] = (0,25 \times -0,10) + (0,50 \times 0,15) + (0,25 \times 0,35) = 13,75\%$
- **Expected Gain (IDR):** $Rp100.000.000 \times 13,75\% = \text{Rp}13.750.000$

13.2 Portfolios

- **Portfolio Weights (w_i):** The percentage of a portfolio's total value invested in a particular asset.
- **Portfolio Expected Return:**

$$E[R_p] = w_1 E[R_1] + w_2 E[R_2] + \cdots + w_n E[R_n]$$

Example (IDR): Total Portfolio: **Rp500.000.000**

- **Stock A:** Rp150.000.000 ($w_A = 0,30$) ; $E[R_A] = 12\%$
- **Stock B:** Rp350.000.000 ($w_B = 0,70$) ; $E[R_B] = 18\%$
- **Portfolio Return:** $(0,30 \times 12\%) + (0,70 \times 18\%) = 16,2\%$

Returns are composed of **Expected** and **Unexpected** parts:

$$R = E[R] + U$$

The unexpected part (U) is driven by **Announcements** (News).

Two Components of Risk

1. **Systematic Risk (m):** Affects many assets (e.g., BI-Rate changes, GDP).
2. **Unsystematic Risk (ϵ):** Affects a single asset (e.g., labor strike at a specific factory).

$$\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}$$

13.5 Diversification and Portfolio Risk

- **The Principle of Diversification:** Spreading an investment across many assets eliminates **unsystematic risk**.
- Diversification can reduce the standard deviation of a portfolio without necessarily lowering expected returns.
- **The Limit:** No matter how many stocks you buy, **systematic risk** cannot be eliminated.

13.6 Systematic Risk and Beta (β)

- **Beta Coefficient:** Measures the amount of systematic risk an asset has relative to an average risky asset (the market).
- $\beta = 1,0 \rightarrow$ Average risk.
- $\beta > 1,0 \rightarrow$ High sensitivity to market (e.g., Tech/Banking).
- $\beta < 1,0 \rightarrow$ Low sensitivity (e.g., Consumer Goods/Utilities).

Portfolio Beta:

$$\beta_p = \sum_{i=1}^{n} w_i \times \beta_i$$

13.7 The Security Market Line (SML)

The SML reflects the relationship between expected return and systematic risk (β).

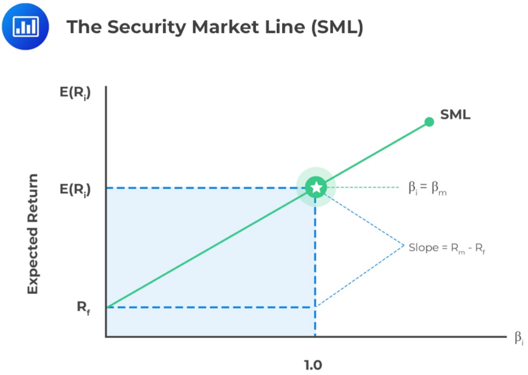
The Capital Asset Pricing Model (CAPM)

$$E[R_i] = R_f + \beta_i \times (E[R_m] - R_f)$$

Variables:

- R_f: Risk-free rate (e.g., 10-year Indonesian Government Bond/SBN).
- $E[R_m] - R_f$: Market Risk Premium.

Visualizing the SML



- The slope of the SML is the **Market Risk Premium**.
- All assets in equilibrium must lie on the SML.

CAPM Example (IDR)

Scenario: You are analyzing a stock in the **IHSG** (Jakarta Composite Index).

- Risk-free rate (R_f): **6,5%** (SBN Yield).
- Market Expected Return ($E[R_m]$): **12,0%**.
- Stock Beta (β): **1,4**.

Calculation:

$$E[R] = 6,5\% + 1,4 \times (12,0\% - 6,5\%)$$

$$E[R] = 6,5\% + 1,4 \times (5,5\%) = 14,2\%$$

If you invest **Rp250.000.000**, your required return is **Rp35.500.000** per year.

13.8 The SML and the Cost of Capital

- **The Basic Idea:** The return an investor *requires* on a risky asset (calculated via CAPM) is the same as the **cost** the firm must pay to use that capital.
- **Investment Criteria:** A project should only be accepted if its expected return is **above** the SML for its level of risk.

Cost of Equity

If a firm has a beta of 1,2, it must earn at least the SML-dictated return to satisfy its shareholders. This is the first step in calculating the **WACC** (Weighted Average Cost of Capital).

Thank You!

Questions & Discussion